

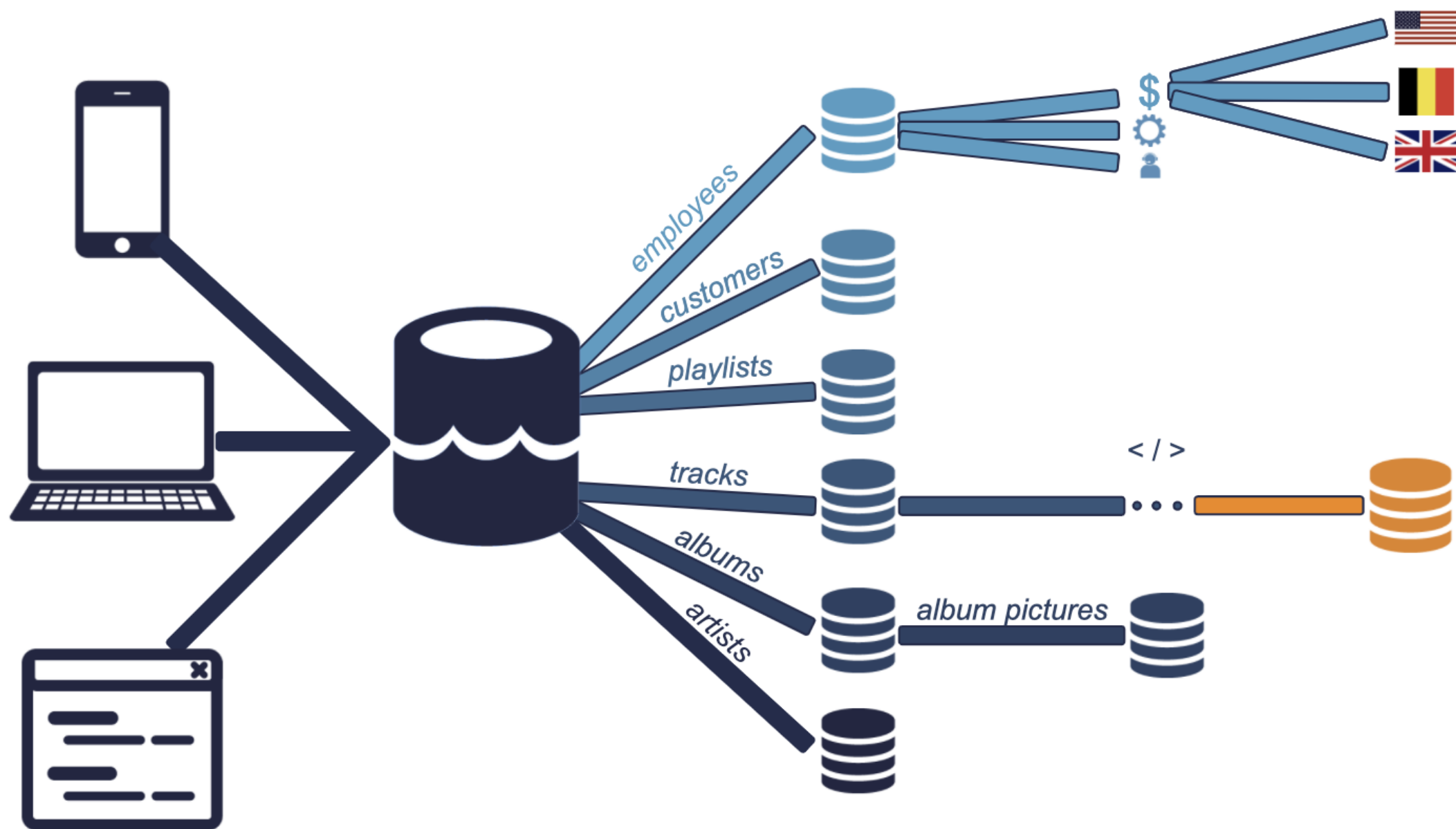
Processing data

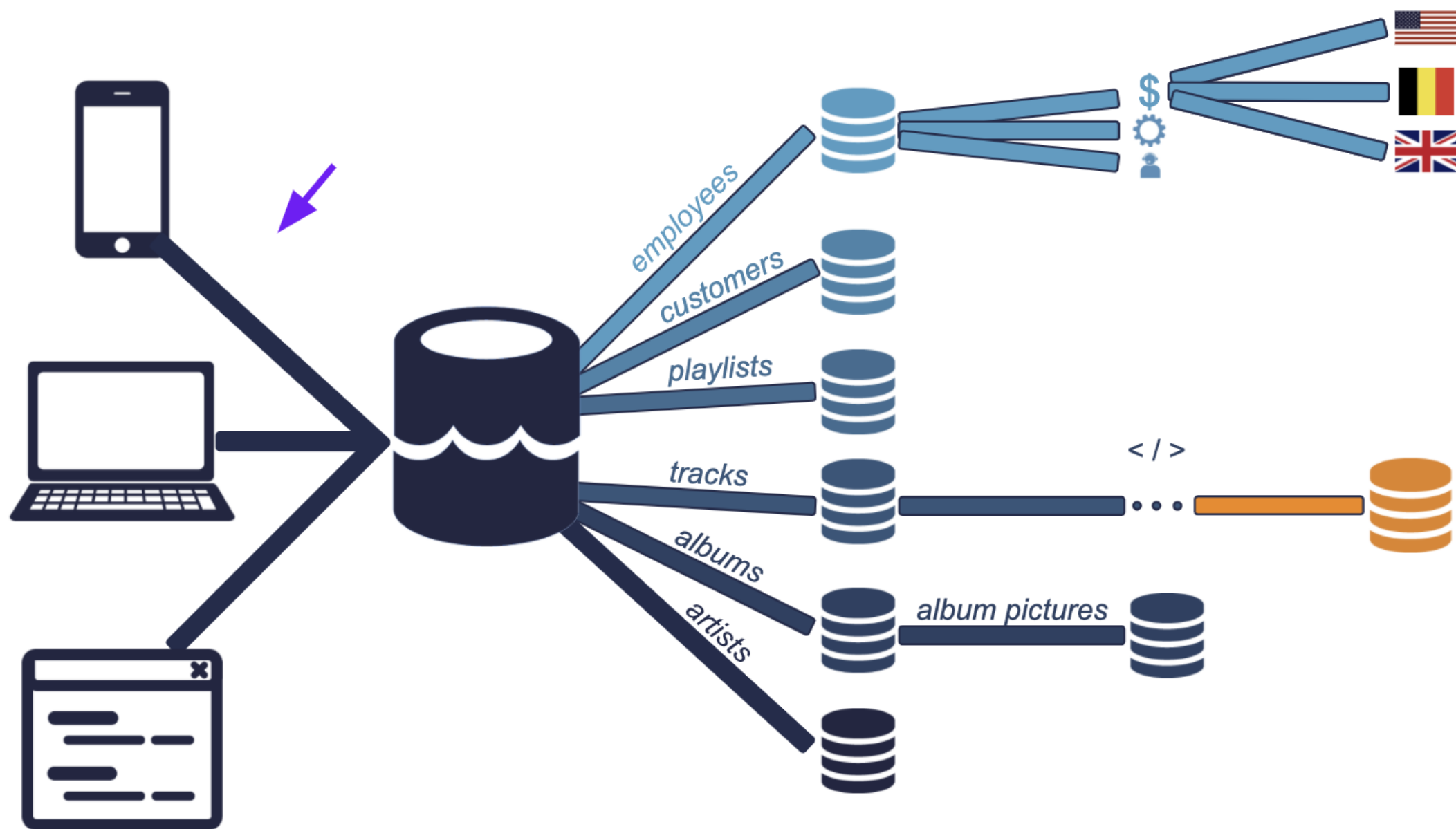
UNDERSTANDING DATA ENGINEERING

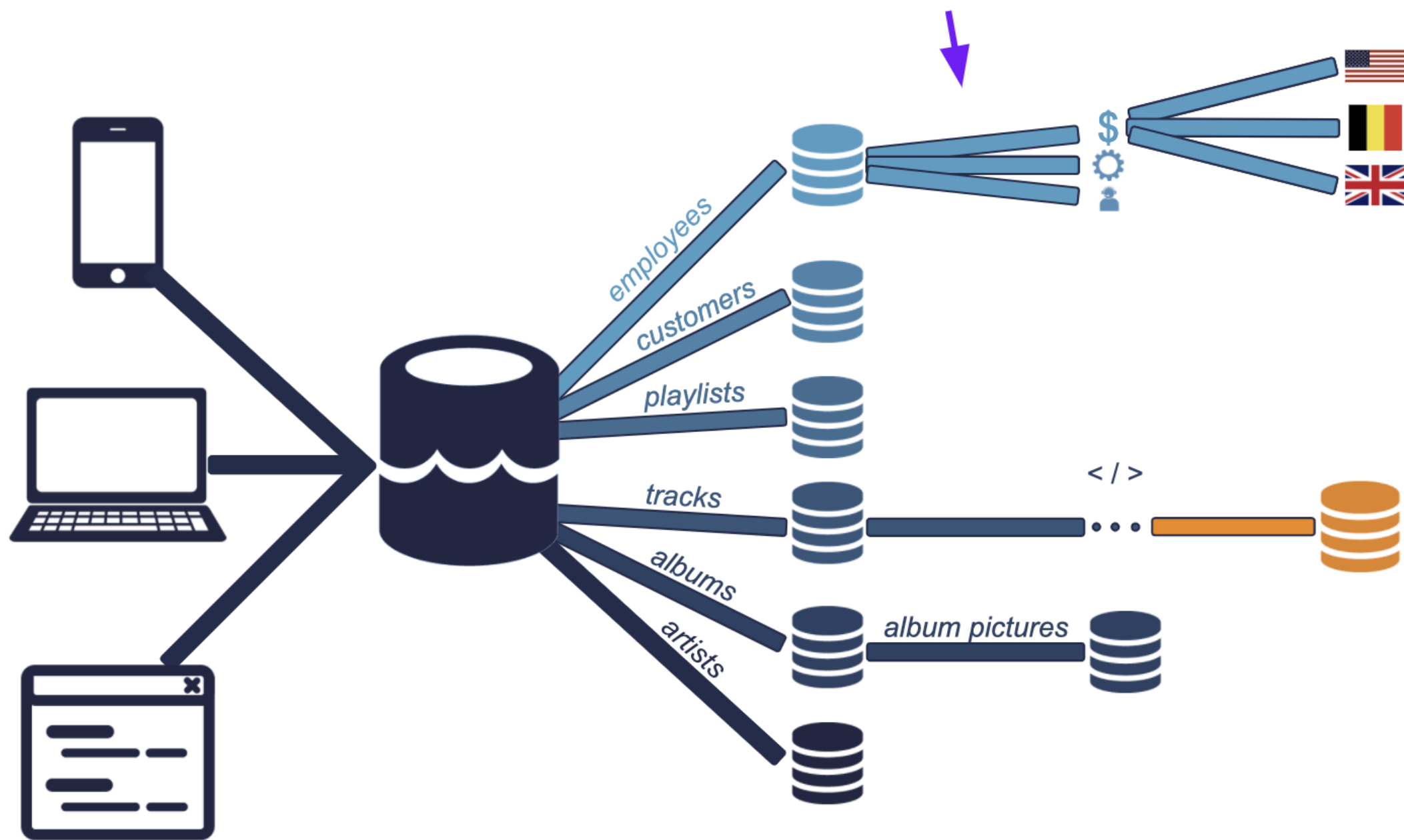


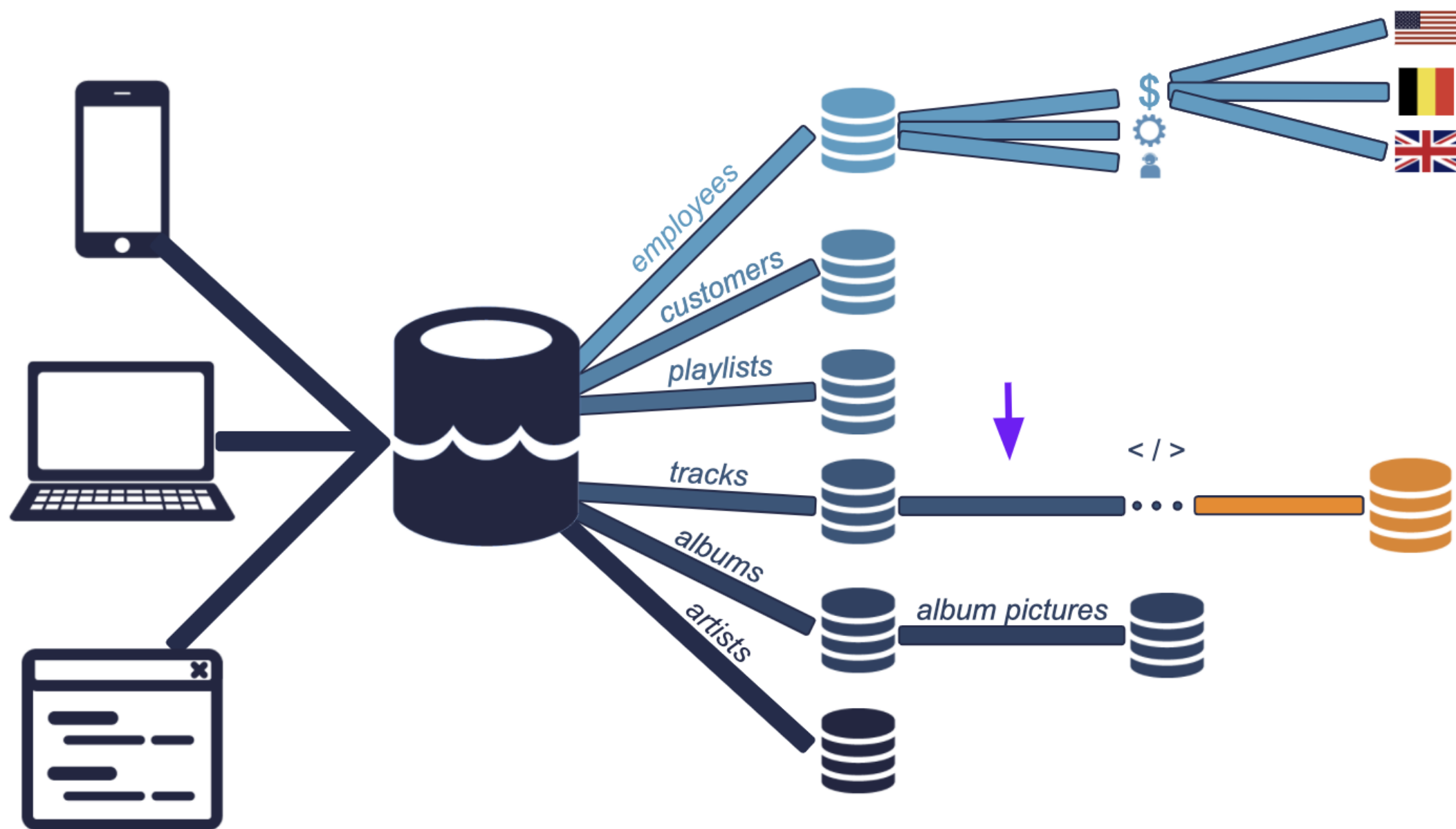
Hadrien Lacroix

Content Developer at DataCamp









A general definition

- Data processing: converting **raw** data into **meaningful** information

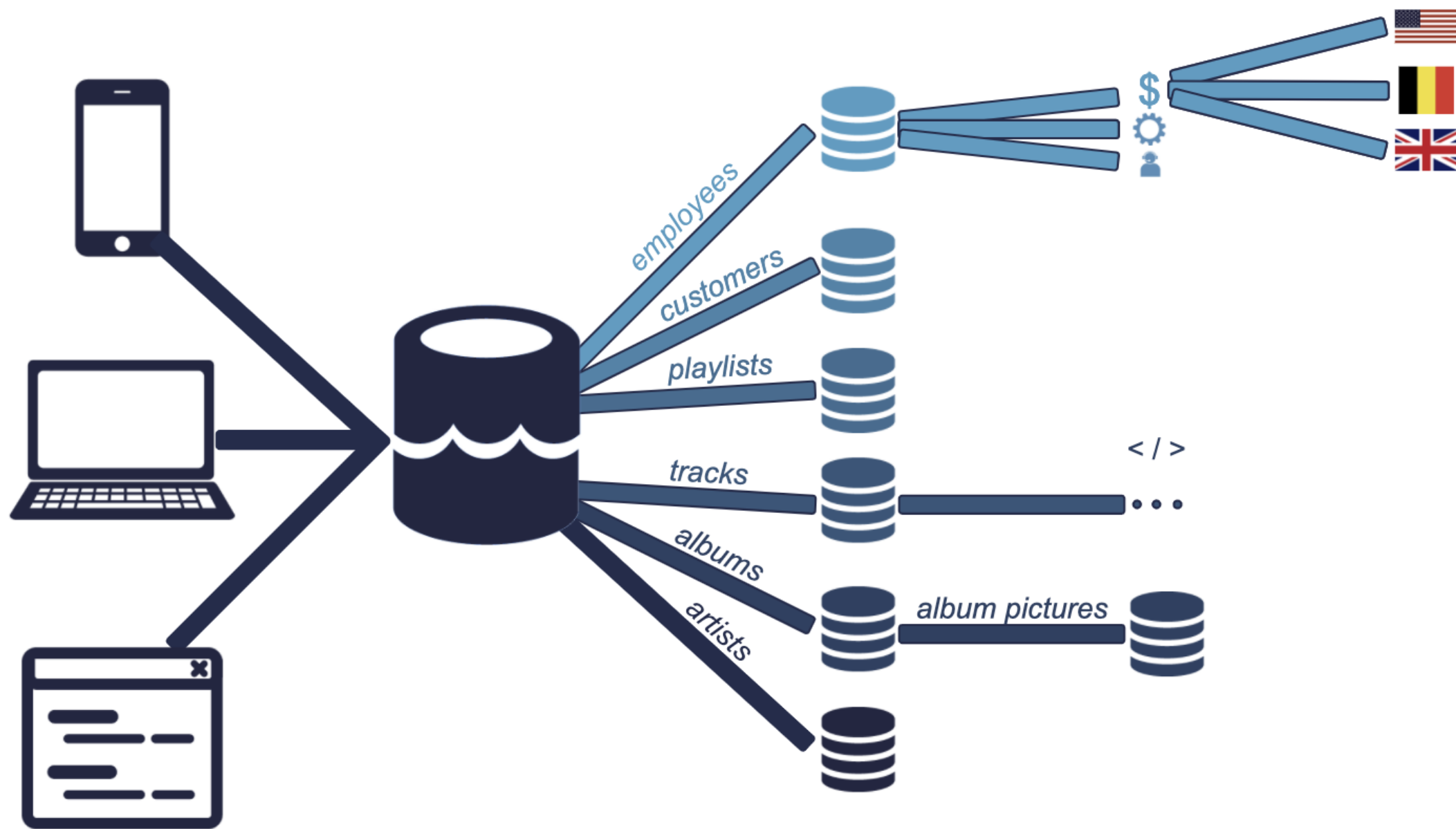
Data processing value

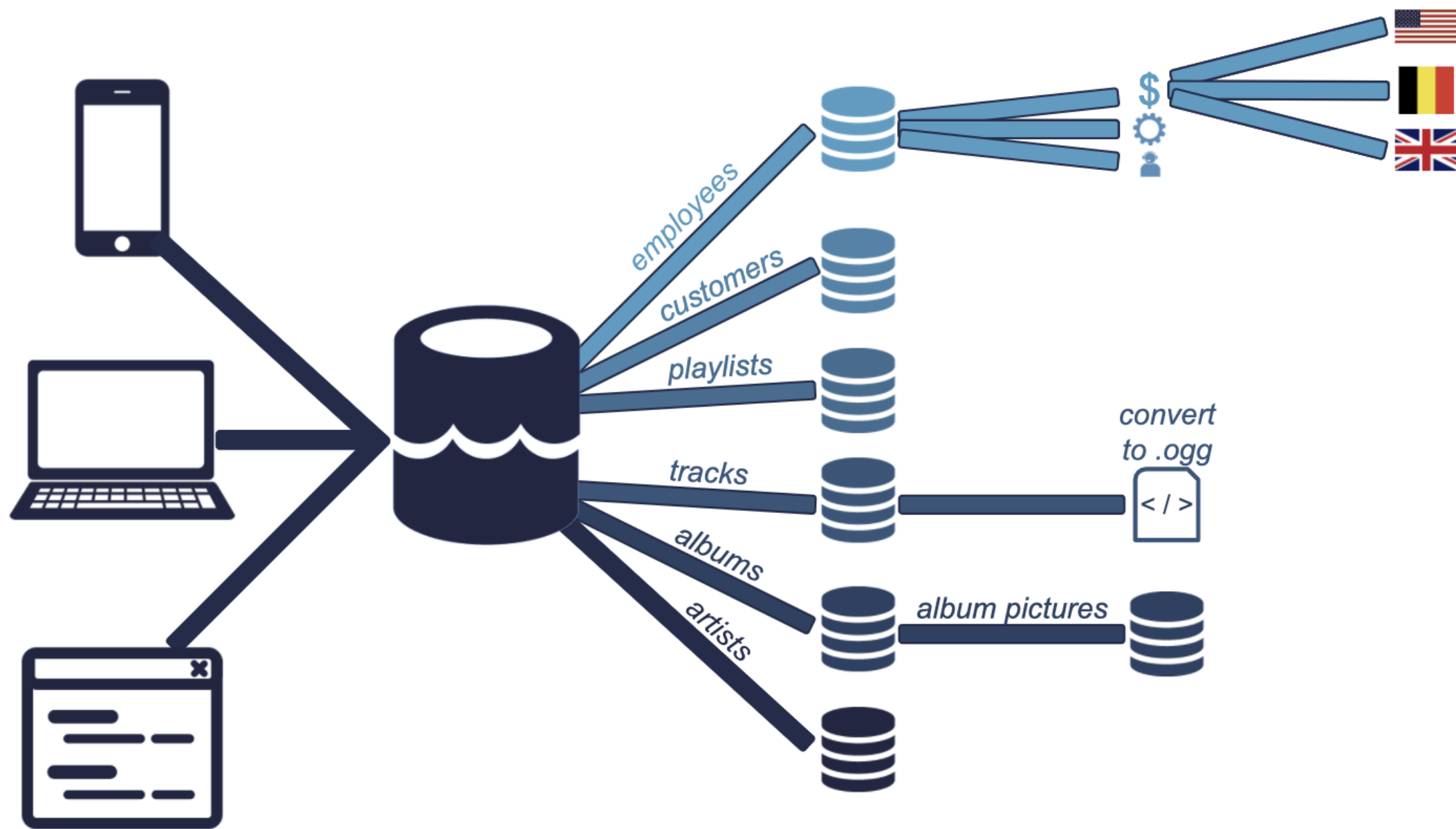
Conceptually

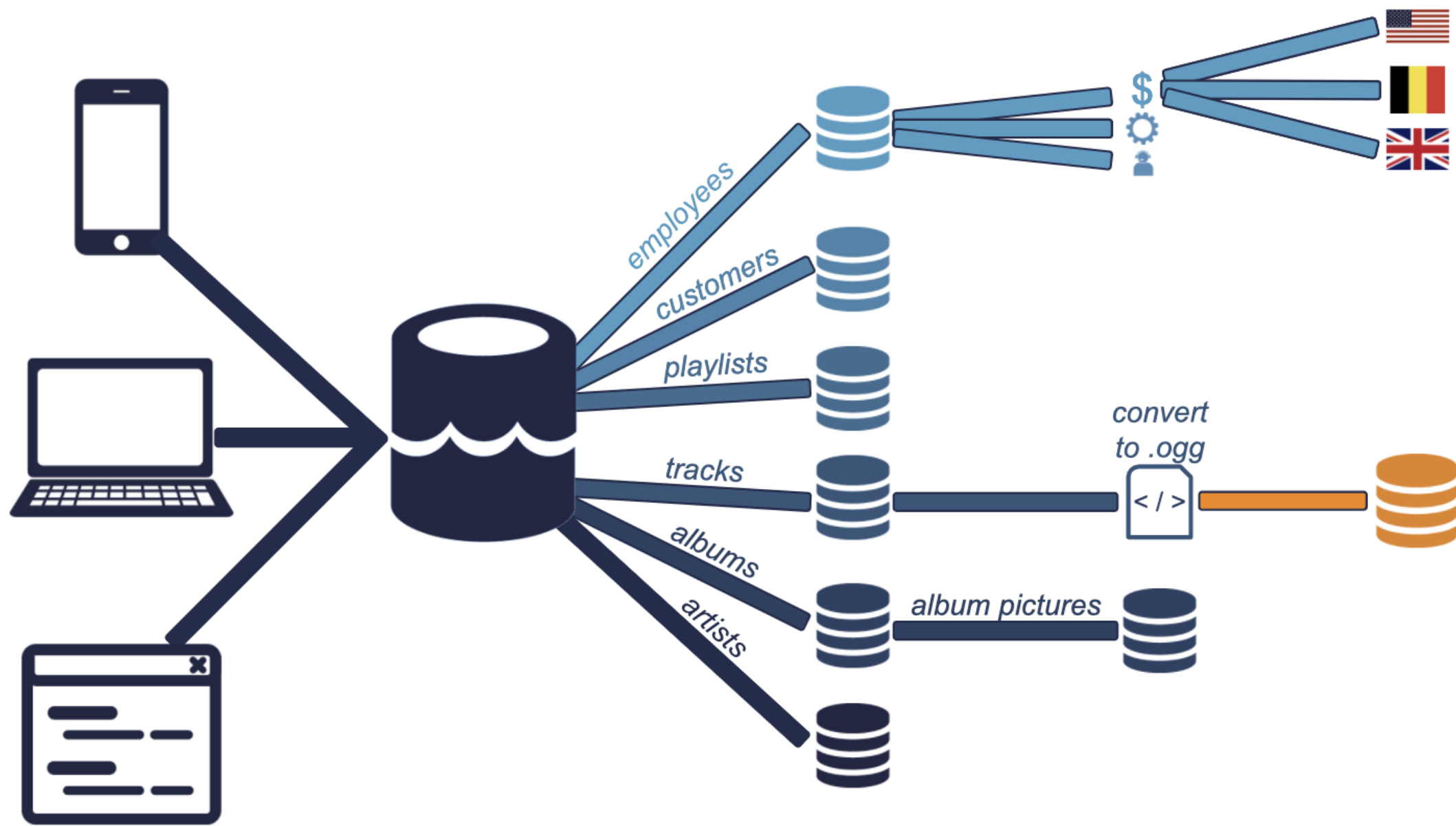
- Remove unwanted data
- Optimize memory, process and network costs
- Convert data from one type to another

At Spotflix

- No long term need for testing feature data
- Can't afford to store and stream files this big







Data processing value

Conceptually

- Remove unwanted data
- To save memory
- Convert data from one type to another
- Organize data
- To fit into a schema/structure
- Increase productivity

At Spotflix

- No need for lossless format
- Can't afford to store files this big
- Convert songs from `.flac` to `.ogg`
- Reorganize data from the data lake to data warehouses
- Employee table example
- Enable data scientists

How data engineers process data

- Data manipulation, cleaning, and tidying tasks
 - that can be automated
 - that will always need to be done
- Store data in a sanely structured database
- Create views on top of the database tables
- Optimizing the performance of the database
- Rejecting corrupt song files
- Deciding what happens with missing metadata
- Separate artists and albums tables...
- ...but provide view combining them
- Indexing

Batch processing



Stream processing



¹ The difference between batch and stream will be explained in the next lesson!



Summary

- What data processing is
- Why it's necessary
- What it consists in
- How we process data at Spotflix

Let's practice!

UNDERSTANDING DATA ENGINEERING

Scheduling data

UNDERSTANDING DATA ENGINEERING



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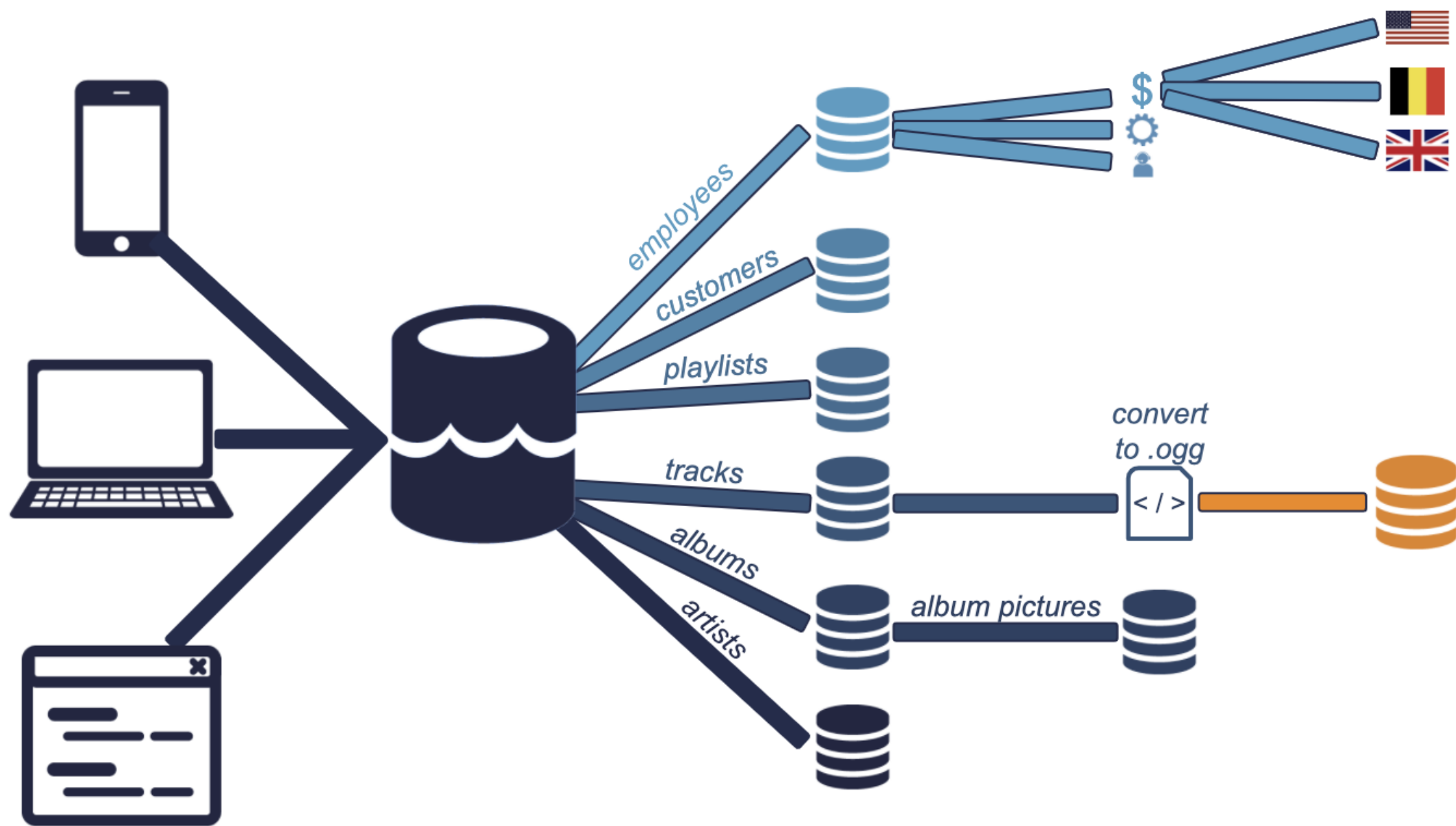
Content Developer at DataCamp

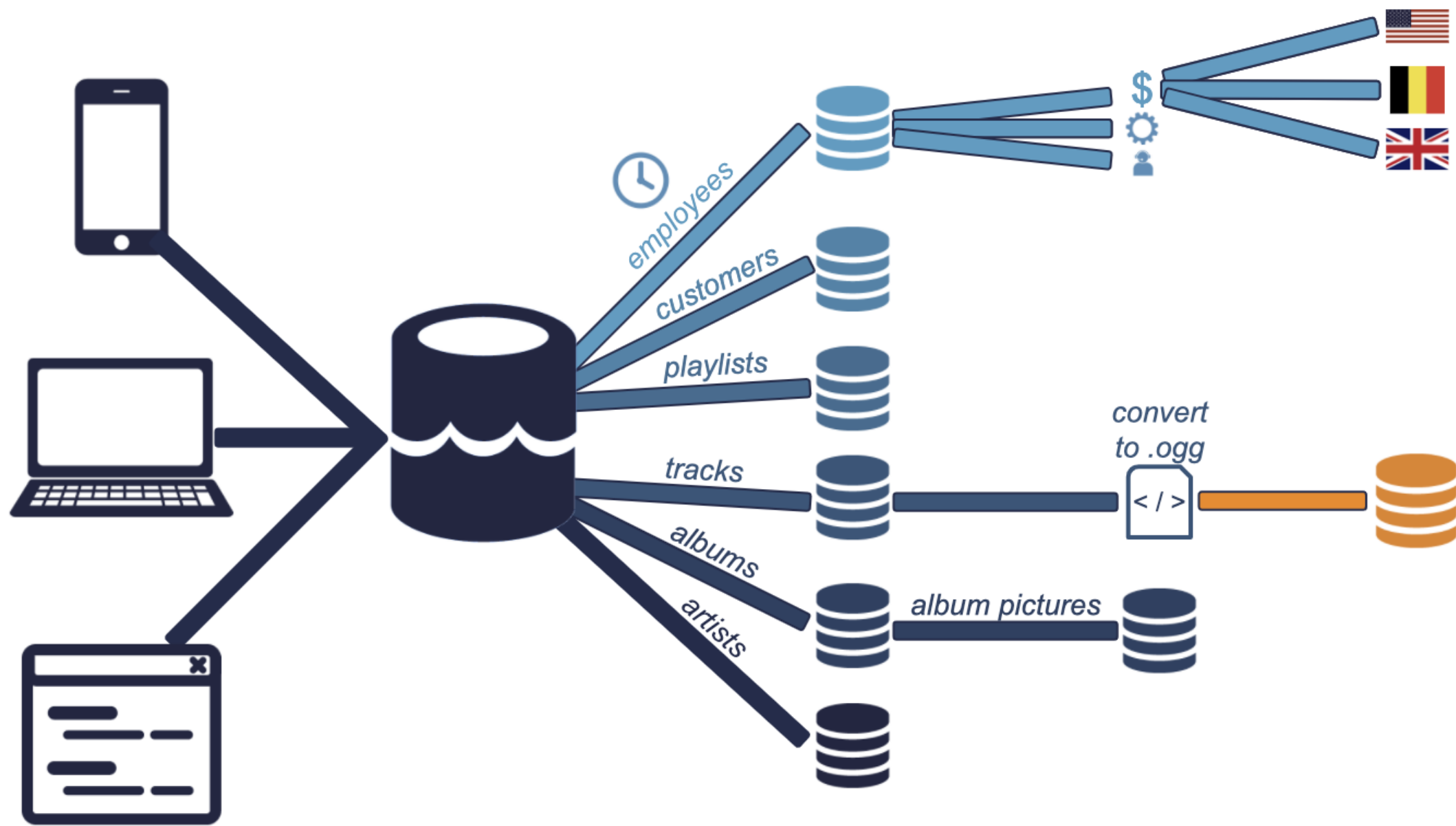
Scheduling

- Can apply to any task listed in data processing
- Scheduling is the glue of your system
- Holds each piece and organize how they work together
- Runs tasks in a specific order and resolves all dependencies

Manual, time and sensor scheduling

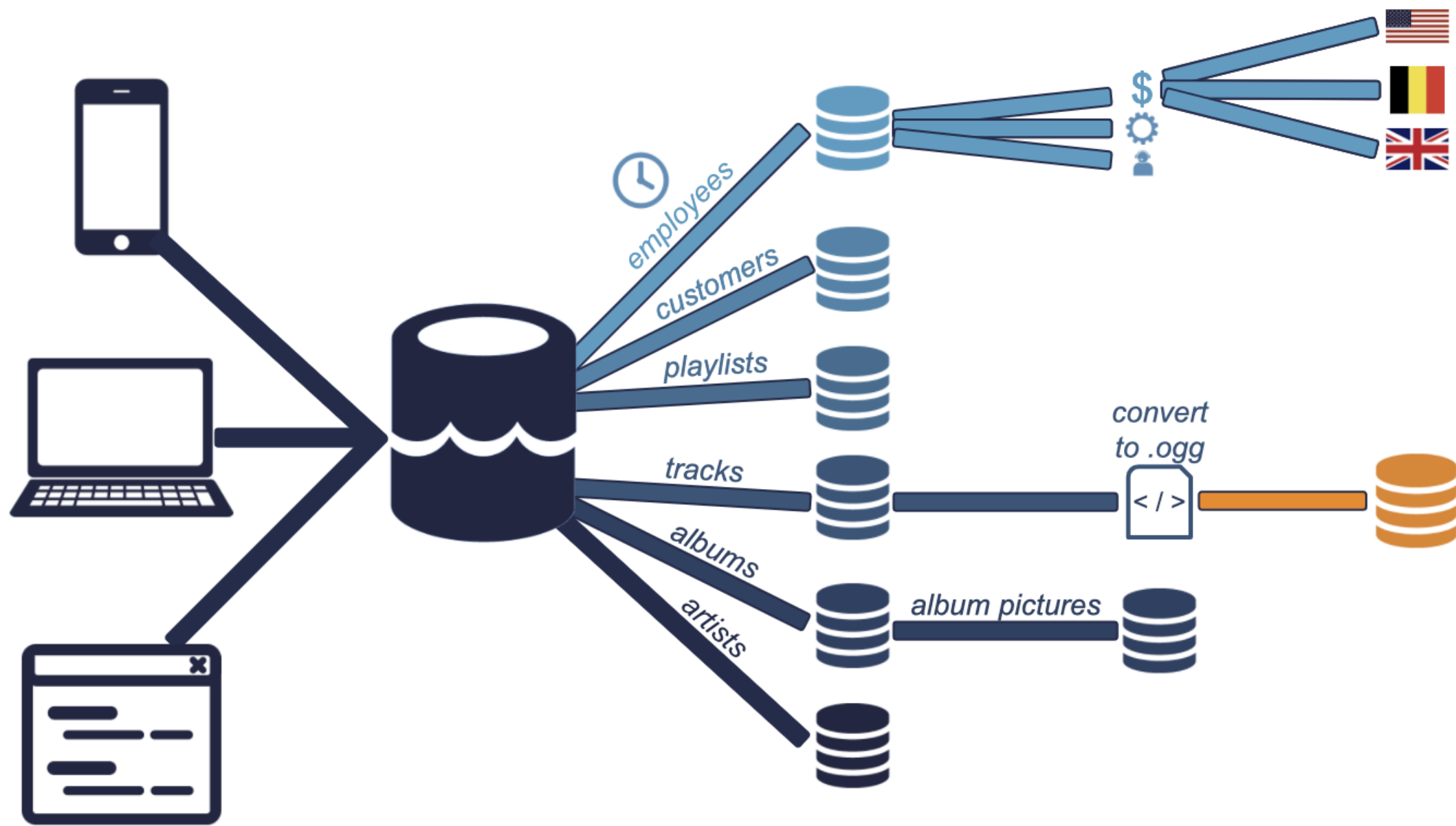
- Manually
- Manually update the employee table

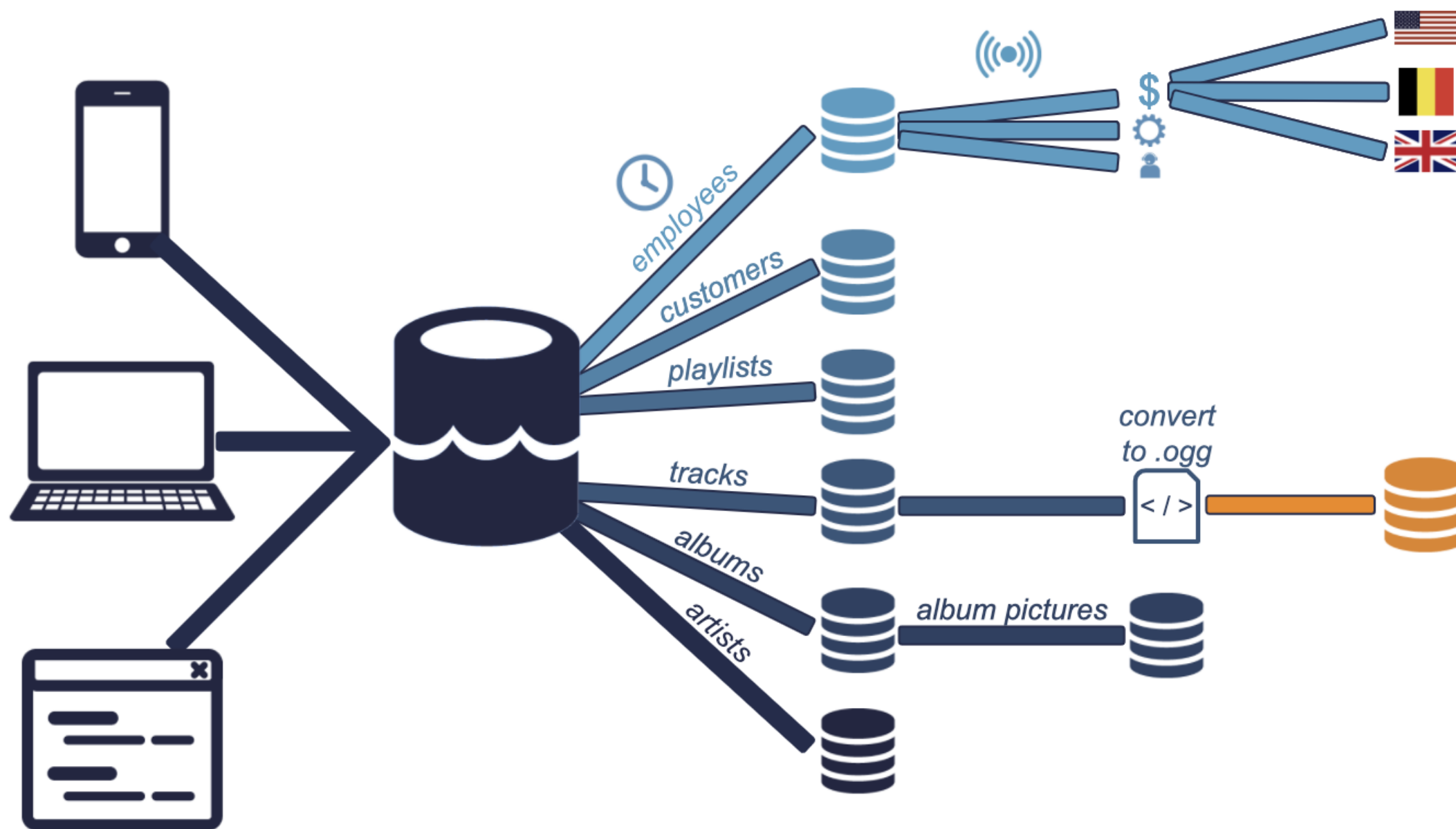




Manual, time and sensor scheduling

- Manually
- Automatically run at a specific time
- Automatically run if a specific condition is met
 - Sensor scheduling
- Manually update the employee table
- Update the employee table at 6 AM





Manual, time, and sensor scheduling

- Manually
- Automatically run at a specific time
- Automatically run if a specific condition is met
 - Sensor scheduling
- Manually update the employee table
- Update the employee table at 6 AM
- Update the department tables if a new employee was added

Batches and streams

- Batches
 - Group records at intervals
 - Often cheaper
 - Streams
 - Send individual records right away
- Songs uploaded by artists
 - Employee table
 - Revenue table
 - New users signing in
 - Another example: online vs. offline listening

Scheduling tools



Summary

- What scheduling is
- Different ways to set it up
- Difference between batches and streams
- How scheduling is implemented at Spotflix
- Airflow, Luigi

Let's practice!

UNDERSTANDING DATA ENGINEERING

Parallel computing

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Parallel computing

- Basis of modern data processing tools
- Necessary:
 - Mainly because of memory
 - Also for processing power
- How it works:
 - Split tasks up into several smaller subtasks
 - Distribute these subtasks over several computers



Time for
100 t-shirts

15



x 1,000

Time for
100 t-shirts

15



x 1,000

30



30



30



30



¹ Emojis by Mohamed Hassan

Time for
100 t-shirts

15



x 1,000

30



30



30



30



Time for
100 t-shirts

15



x 1,000

30



x 250

30



x 250

30



x 250

30



x 250

Time for
100 t-shirts

15



x 1,000

Time for 1,000 t-shirts

30



x 250

1h15



30



x 250



30



x 250



30



x 250



Time for
100 t-shirts

15



x 1,000

Time for 1,000 t-shirts
2h30

15

15

15

15

15

15

15

15

15

15

15



30



x 250

1h15

30

30

15



30



x 250

30

30

15



30



x 250

30

30

15



30



x 250

30

30

15



Benefits and risks of parallel computing

- Employees = processing units
- Advantages
 - Extra processing power
 - Reduced memory footprint
- Disadvantages
 - Moving data incurs a cost
 - Communication time

Time for
100 t-shirts

15



x 1,000

Time for 1,000 t-shirts
2h30

15 15 15 15 15 15 15 15 15 15



30



x 250

1h15

30 30 15



30



x 250

30 30 15



30



x 250

30 30 15



30



x 250

30 30 15



Time for
100 t-shirts

15



x 1,000

Time for 1,000 t-shirts
2h30

15 15 15 15 15 15 15 15 15 15



30



0h10
x 250

1h15

30 30 15



30



x 250

30 30 15



30



x 250

30 30 15



30



x 250

30 30 15



Time for
100 t-shirts

15



x 1,000

Time for 1,000 t-shirts
2h30

15

15

15

15

15

15

15

15

15

15



30



0h10

x 250

1h15

30

30

15

0h05



30



x 250

30

30

15



30



x 250

30

30

15



30



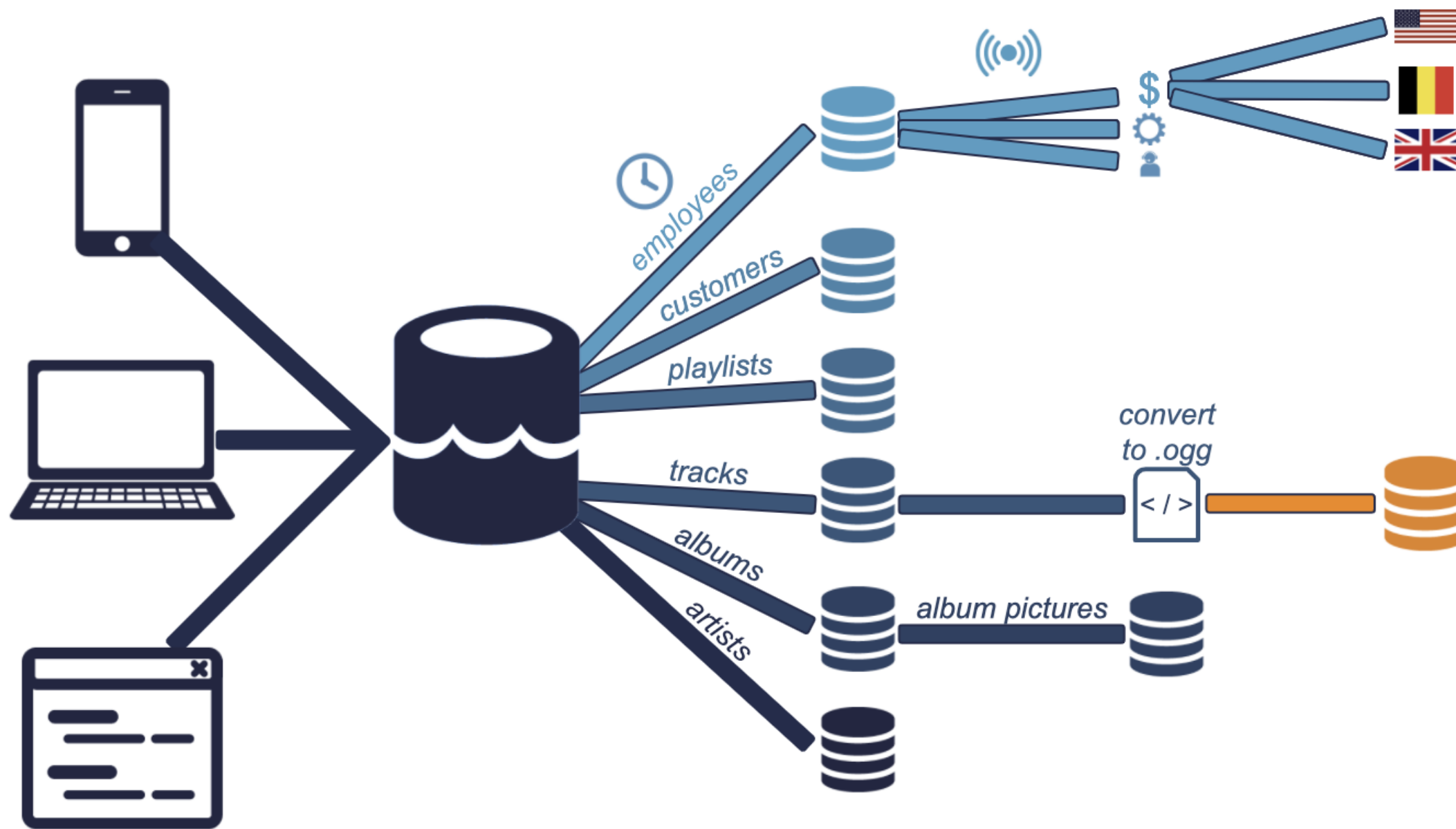
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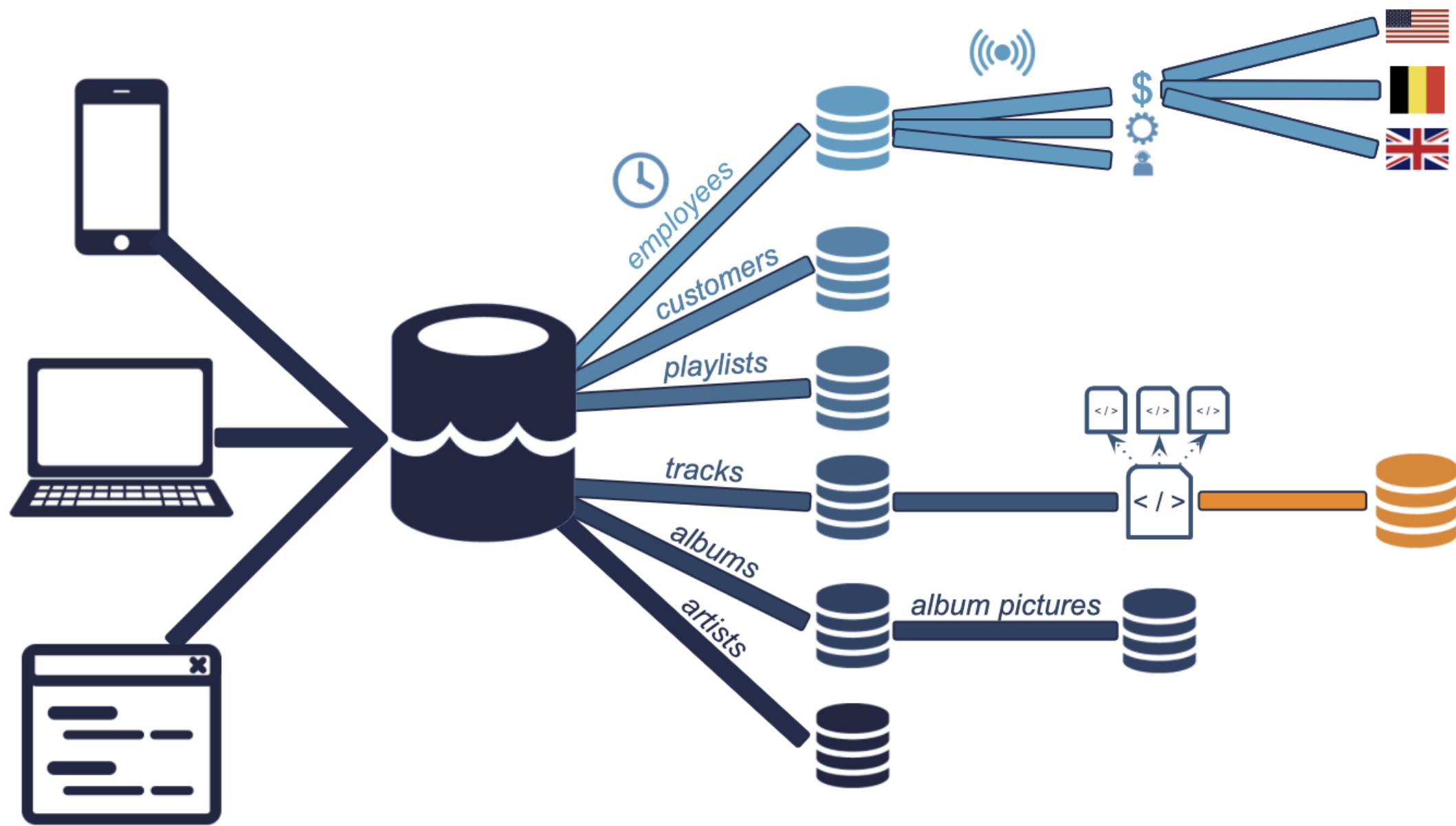
30

30

15







Summary

- Benefits and risks
- How it's implemented at Spotflix

Let's practice!

UNDERSTANDING DATA ENGINEERING

Cloud computing

UNDERSTANDING DATA ENGINEERING



Hadrien Lacroix
Content Developer

Cloud computing for data processing

Servers on premises

- Bought
- Need space
- Electrical and maintenance cost
- Enough power for peak moments
- Processing power unused at quieter times

Servers on the cloud

- Rented
- Don't need space
- Use just the resources we need
- When we need them
- The closer to the user the better

Cloud computing for data storage

- Database reliability: data replication
- Risk with sensitive data

aws
32.4%



32.4%



17.6%



32.4%



17.6%



6%

File storage



File storage

AWS S3



File storage

AWS S3



Azure
Blob Storage



File storage

AWS S3



Azure
Blob Storage



Google
Cloud Storage



File storage

Computation



AWS S3



*Azure
Blob Storage*



*Google
Cloud Storage*





File storage

AWS S3



Computation

AWS EC2



*Azure
Blob Storage*



*Google
Cloud Storage*





File storage

AWS S3



Computation

AWS EC2



*Azure
Blob Storage*



*Azure
Virtual Machines*



*Google
Cloud Storage*





File storage

AWS S3



*Azure
Blob Storage*



*Google
Cloud Storage*



Computation

AWS EC2



*Azure
Virtual Machines*



*Google
Compute Engine*





File storage

AWS S3



Computation

AWS EC2



Databases



*Azure
Blob Storage*



*Azure
Virtual Machines*



*Google
Cloud Storage*



*Google
Compute Engine*





File storage

AWS S3



*Azure
Blob Storage*



*Google
Cloud Storage*



Computation

AWS EC2



*Azure
Virtual Machines*



*Google
Compute Engine*



Databases

AWS RDS





File storage

AWS S3



*Azure
Blob Storage*



*Google
Cloud Storage*



Computation

AWS EC2



*Azure
Virtual Machines*



*Google
Compute Engine*



Databases

AWS RDS



*Azure
SQL Database*





File storage

AWS S3



Azure
Blob Storage



Google
Cloud Storage



Computation

AWS EC2



Azure
Virtual Machines



Google
Compute Engine



Databases

AWS RDS

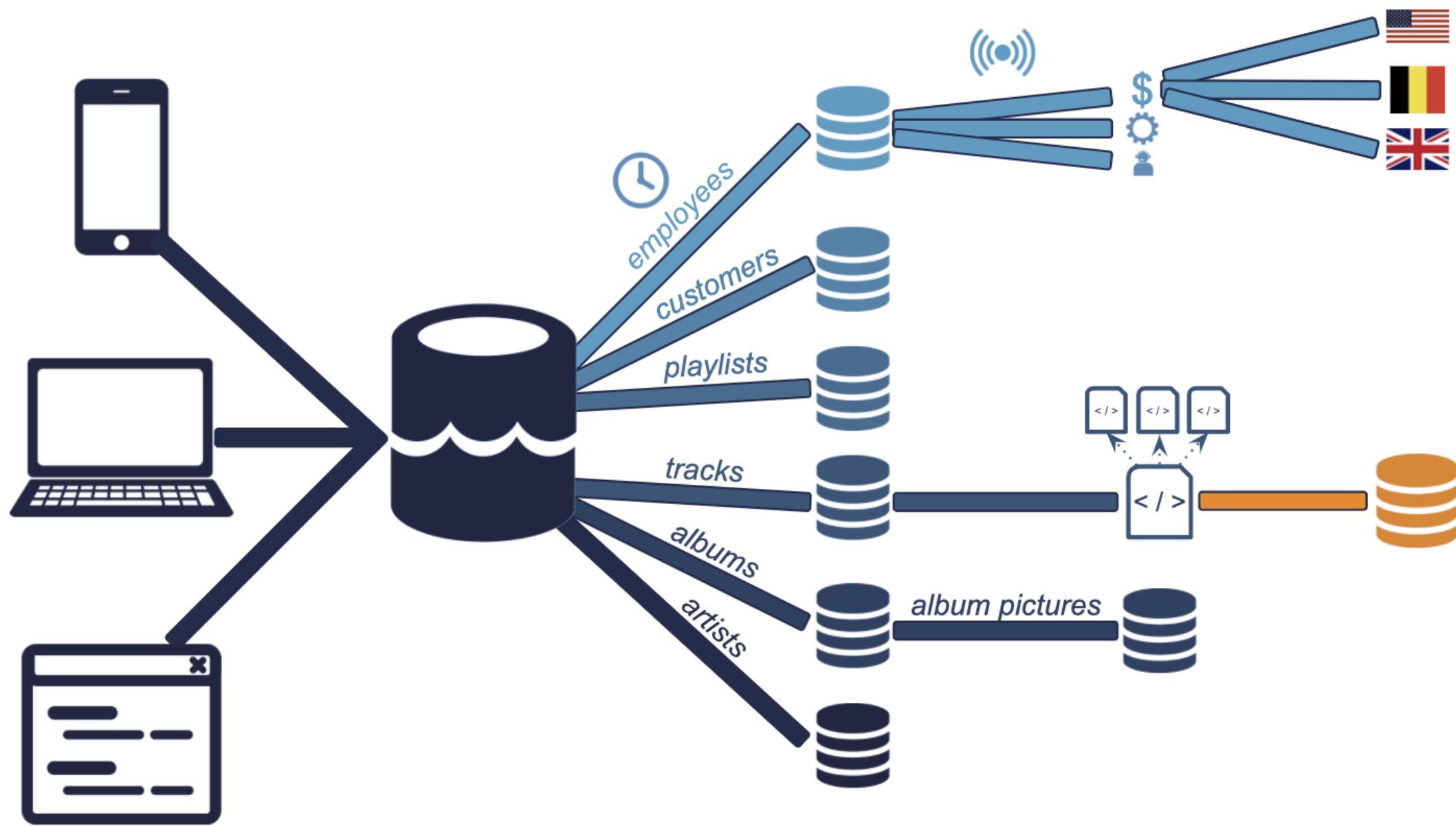


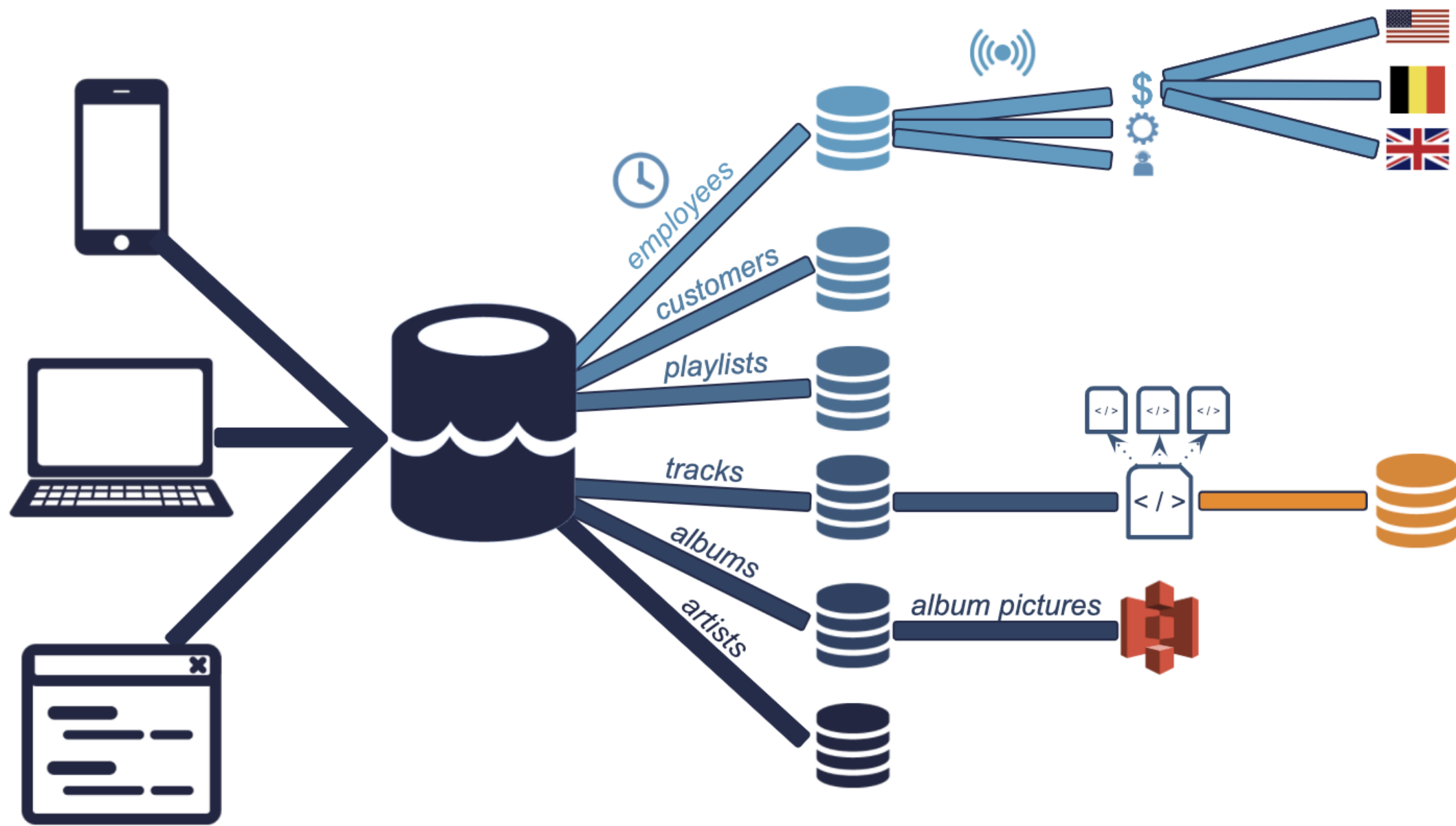
Azure
SQL Database

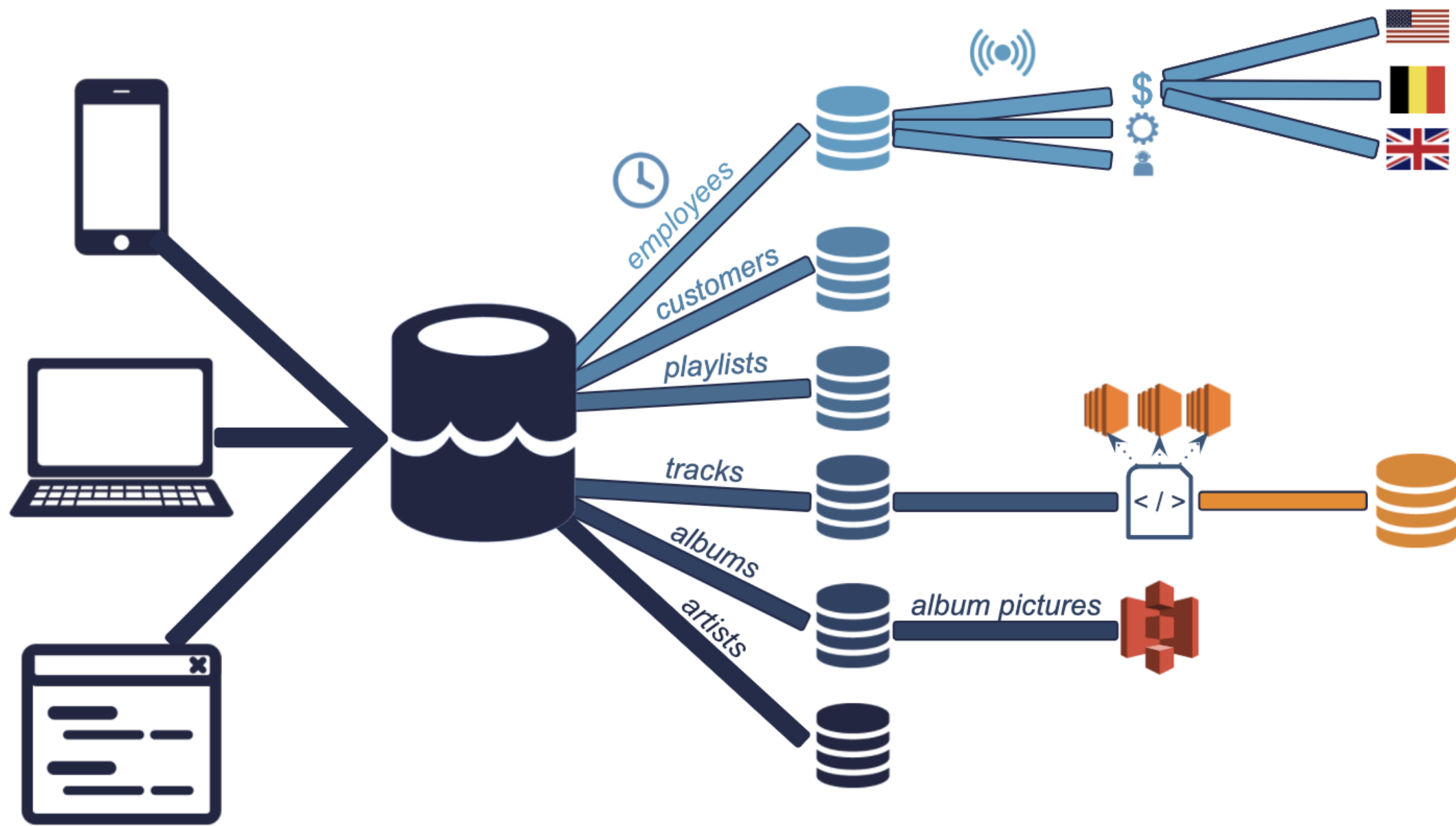


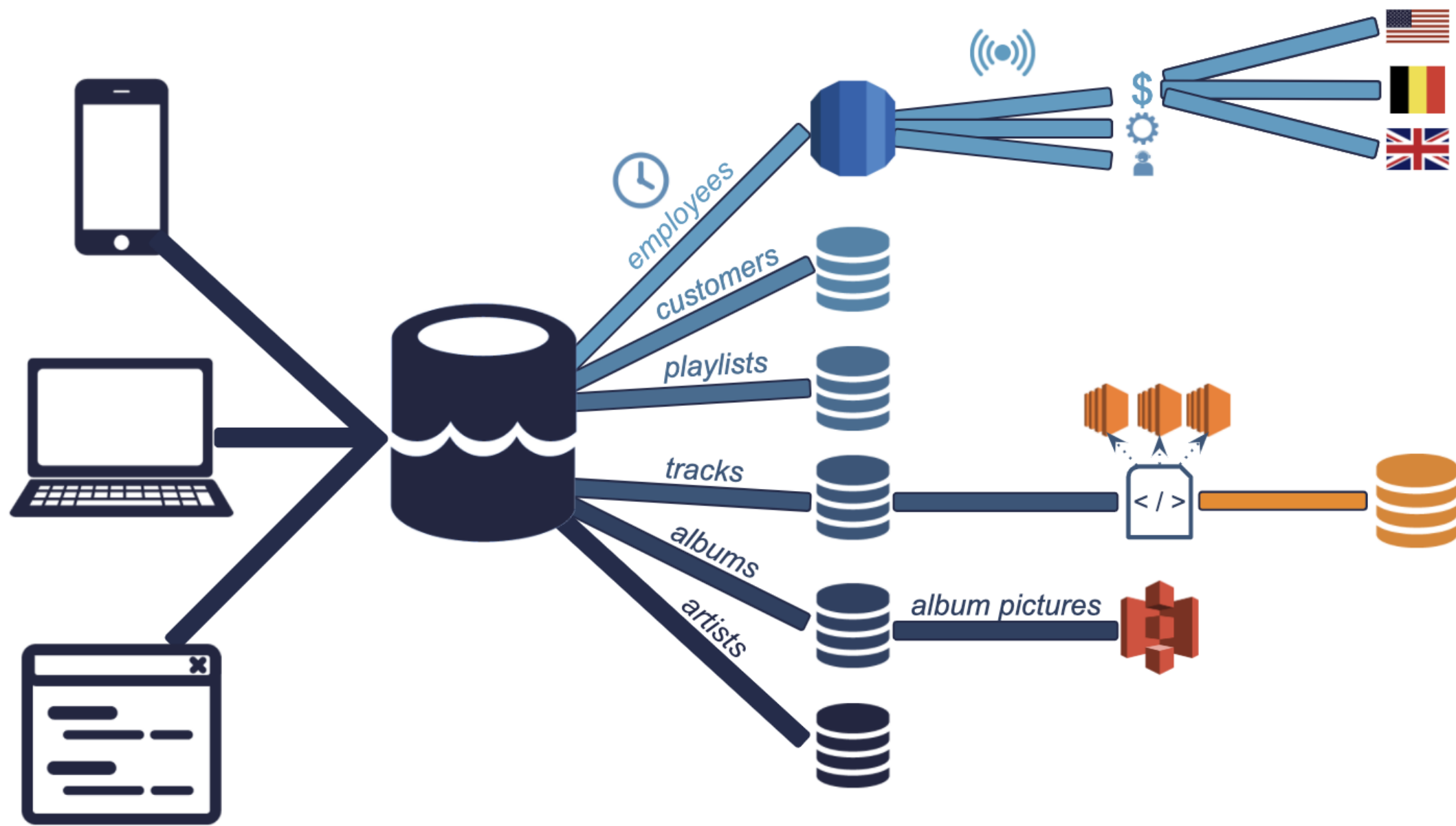
Google
Cloud SQL











Multicloud

Pros

- Reducing reliance on a single vendor
- Cost-efficiencies
- Local laws requiring certain data to be physically present within the country
- Mitigating against disasters

Cons

- Cloud providers try to lock in consumers
- Incompatibility
- Security and governance

Summary

- Benefits and risks of cloud computing
- How it is implemented at Spotflix
- Can cite the main cloud providers and their services

Let's practice!

UNDERSTANDING DATA ENGINEERING

We are the champions

UNDERSTANDING DATA ENGINEERING



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Content Developer at DataCamp

Actually, YOU are the champion!



What you learned - chapter 1

- What Data Engineering is
- How important it is
- How data engineers differ from data scientists
- What a data pipeline is and how it works

What you learned - chapter 2

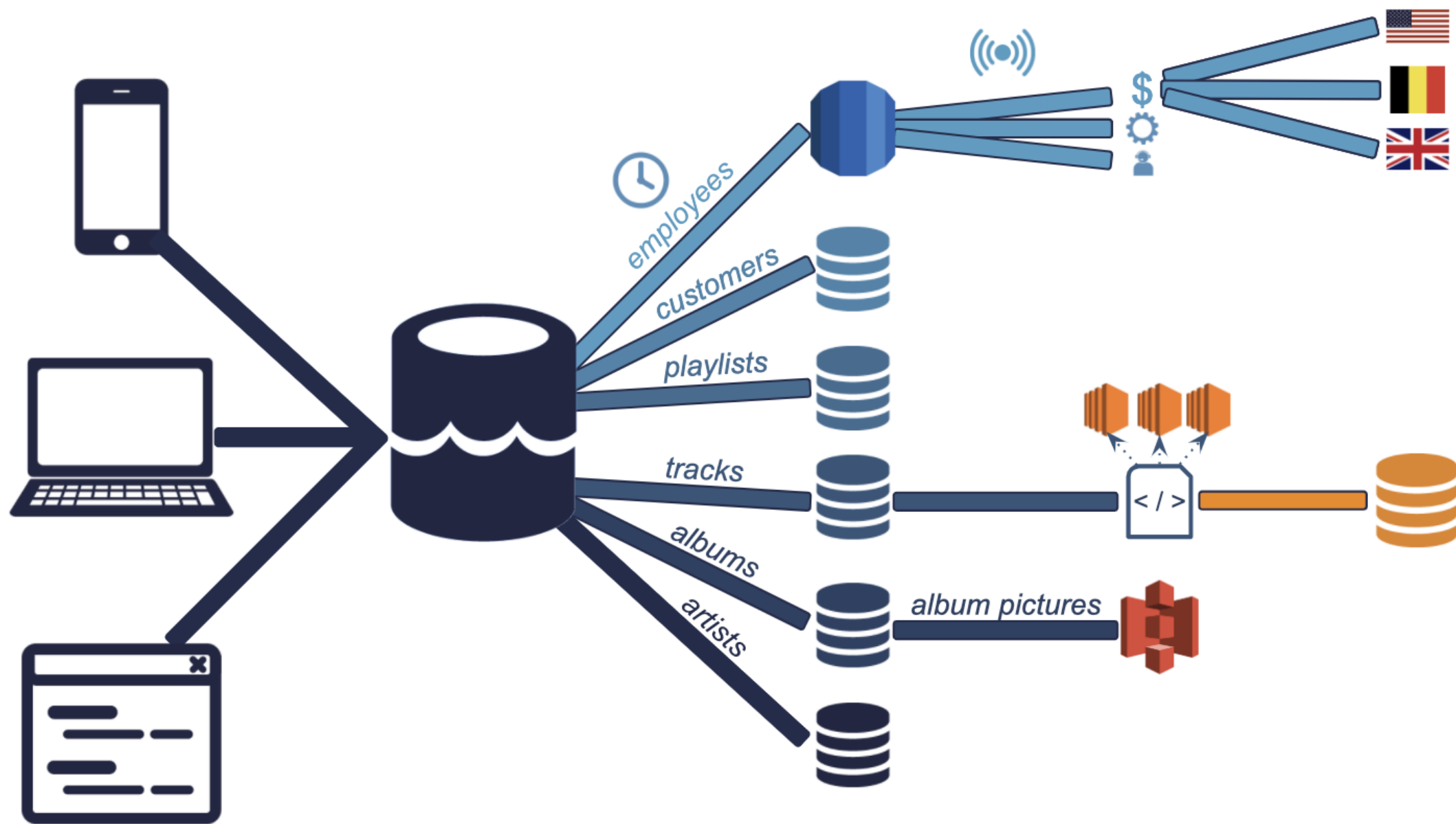
- The different structures data can take
- How fundamental SQL is
- The differences between data lakes, data warehouses and databases

What you learned - chapter 3

- How data is processed
- How scheduling holds it all together
- Parallel computing
- Cloud computing

And some more

- What SQL code actually looks like
- Main tools and technologies used in data engineering
- And some more



Data Engineering for Everyone - Lexicon

Data Engineering for Everyone - Lexicon

- **Airflow**: an open-source workflow management platform used to schedule data engineering tasks. Started at Airbnb, now maintained by the Apache foundation.
- **AWS**: Amazon Web Services. Amazon's cloud computing services.
- **Azure**: Microsoft's cloud services.
- **Big data**: the systematic storage, management and analysis of datasets that are too large or complex to be dealt with by traditional data-processing application software. Big Data revolves around 4 Vs: volume, variety, velocity, and veracity.
- **Cloud computing**: the use of a network of remote servers hosted on the Internet to store, manage, and process data, rather than a local server or a personal computer.

A promise is a promise, DataChamps!

- All the exercises are song titles
- Search for "DataChamps" on Spotify

Congratulations!

UNDERSTANDING DATA ENGINEERING